

Plexo: Singapore School Examination Seating Platform

Official User Guide & Operations Manual (Version 0.1)

1. Product Purpose & The Local-Only Workflow Rationale

1.1 Purpose of the Product

Plexo is a specialized, zero-backend examination planning, seating optimization, and audit engine designed specifically for the operational realities of Singapore primary and secondary schools administering **SEAB National Examinations** (GCE N(T), N(A), O-Level, and A-Level) as well as internal school preliminary examinations.

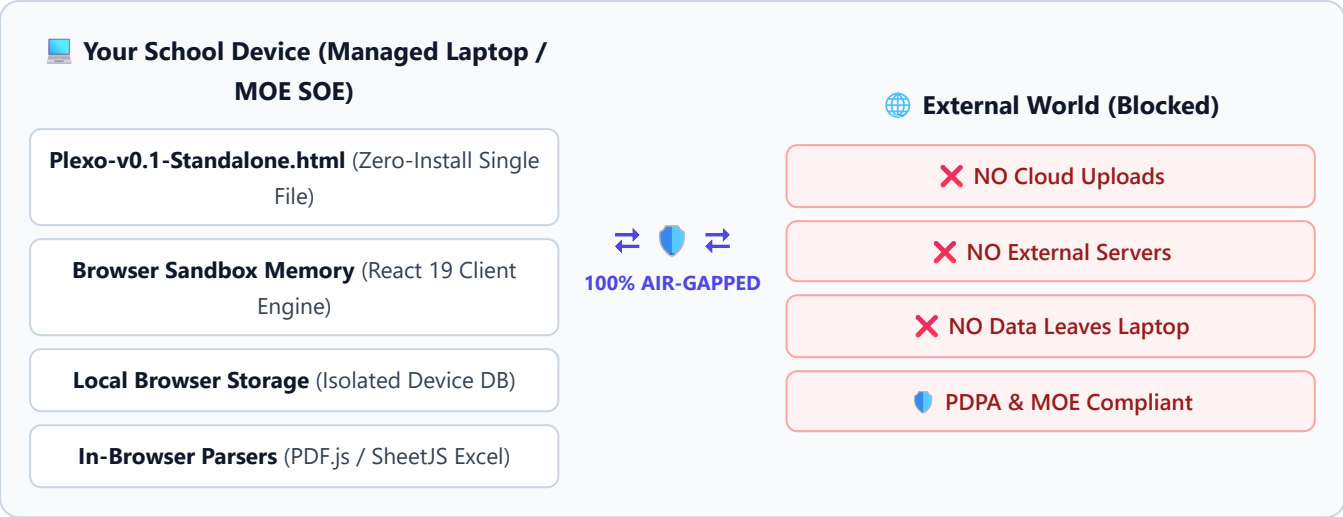
School examination operations require complex logistics:

- **Consecutive Paper Continuity:** Candidates sitting for back-to-back papers (e.g., English Paper 1 followed by Paper 2 with a 45-minute recess) must retain their designated desks to eliminate hall confusion.
- **Granular Access Arrangements (AA):** Accommodating candidates requiring extra time (15%, 25%, 50%), quiet segregated rooms, or preferential front-row seating for Listening Comprehension—often varying on a paper-by-paper basis.
- **Multi-Shift Laboratory & PC Allocations:** Science practicals (Physics, Chemistry, Biology) and Computing examinations with shift rotations and holding rooms.
- **Audit & Compliance Documentation:** Generating SEAB-compliant door noticeboard cards, 8-up candidate desk slips, daily room schedule matrices, and invigilator attendance rosters.

Plexo replaces vulnerable, manual Excel spreadsheets with an automated, deterministic solver that completes whole-cohort seating allocations in seconds.

1.2 The Critical Importance of a 100% Fully Local Workflow

Examination management involves sensitive student demographic, academic, and medical data. Plexo is intentionally engineered with a **zero-backend, client-side-only architecture** to uphold the highest data governance standards:



⚠ IMPORTANT: DATA PRIVACY & COMPLIANCE

Data Privacy & PDPA Compliance (Zero Data Egress):

- **Zero Remote Transmission:** All file parsing, algorithmic seat solving, 2D floorplan rendering, and PDF report generation occur **100% inside your browser's local sandbox**.
- **Immediate NRIC/FIN Sanitization:** Upon file upload, national identification numbers (NRIC/FIN) are parsed strictly to verify uniqueness and **immediately discarded from memory**. Candidates are tracked purely by their statutory 4-digit Index Number (e.g., 0005).
- **Confidential AA Protection:** Medical, psychological, and mobility accommodation records remain strictly on the physical laptop running the session.
- **Managed School Laptops (MOE SOE):** Works directly by double-clicking `Plexo-v0.1-Standalone.html`. Requires **no admin rights, no server installation, no Node.js, and no Python**.

2. Required Input Files

To configure your school's examination session, prepare the following two official files:

File Type	Accepted File Formats	Source / Description	Key Data Extracted
1. Official SEAB Candidate List	.xlsx , .xls , .csv	Exported directly from the school administration cockpit or SEAB registration portal. Multi-file upload is supported.	- Academic Level - Statutory Full Name 4-Digit Index Number - Enrolled Subject / Paper Codes (e.g., 1128/01 , 6091/01)
2. Official SEAB Exam Timetable	.pdf , .xlsx , .xls , .csv	The official PDF calendar published by SEAB (e.g., GCE O-Level Timetable) or school internal spreadsheet.	- Examination Date (YYYY-MM-DD) - Start Time (HH:mm) - Duration (minutes) - Paper Title & Type (WRITTEN , SCIENCE_LAB , LISTENING_COMP)

💡 OPERATIONAL TIP

Multi-File Upload: You can drag-and-drop multiple candidate files simultaneously (e.g., class-by-class spreadsheets: 4E1.csv , 4E2.csv , 4N1.csv). Plexo automatically merges enrollments and indexes candidates by index number with zero duplication.

3. Step-by-Step User Guide

Step 0: Launching Plexo on a Managed Laptop

1. Open your File Explorer.
2. Locate Plexo-v0.1-Standalone.html (or extract plexo-v0.1.zip).
3. **Double-click** Plexo-v0.1-Standalone.html . It opens directly in Microsoft Edge or Google Chrome under file:///... .
4. Observe the primary navigation bar across the top displaying the 5 sequential operational stages:

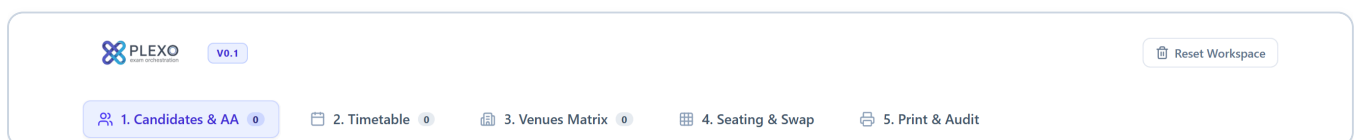


Figure: Plexo Application Header & Navigation Bar

Step 1: Stage 1 — Candidate Directory & Access Arrangements (AA)

1.1 Ingesting Candidate Data

- Click **Stage 1: Candidates & AA**.

- Click **"Upload Files (.xlsx / .csv)"** or drag your registration files directly onto the ingestion zone.
- You may also click **"Sample Template"** to download an authentic SEAB-formatted reference CSV.

Stage 1: Candidate Directory & AA Registry

Upload single or multiple SEAB candidate registration lists (multi-file supported). NRIC/FIN is discarded; last 4 digits of Index No. serve as Candidate ID.

Sample Template

Upload Files (.xlsx / .csv)

Figure: Stage 1 Candidate Ingestion Banner

1.2 Inspecting the Candidate Roster

Once uploaded, Plexo calculates summary tallies across the cohort (Standard Seating, Total AA Flagged, Extra Time, Separate Room, and Front Row Preferential Seating).

- Candidates are ordered by their 4-digit Index Number.
- Enrolled subject codes are listed as clickable badges.
- Any active Access Arrangements are displayed as color-coded tags.

Stage 1: Candidate Directory & AA Registry

Upload single or multiple SEAB candidate registration lists (multi-file supported). NRIC/FIN is discarded; last 4 digits of Index No. serve as Candidate ID.

Sample Template

Upload Files (.xlsx / .csv)

Figure: Stage 1 Candidate Roster & AA Metrics

1.3 Configuring Granular Paper-Level Access Arrangements (AA)

Examinations often require accommodations that differ by subject (e.g., +25% extra time for English Writing, but standard conditions for Mathematics).

1. Click **" + Add AA "** or **"Edit AA "** on any candidate row.
2. **"All Papers (Default)" Tab:** Sets baseline accommodations (+15%, +25%, +50%, Separate Room, Preferential Front Row, and Clinical Remarks).
3. **Individual Paper Tabs (e.g., 1128/01):** Click any registered paper tab to customize accommodations exclusively for that paper.
4. Click **"Save Arrangements"**. Overridden papers display a purple badge on the candidate row.

Access Arrangement (AA) Configuration

SUBJECT & PAPER GRANULAR

×

Candidate: 0002 • LIM XIAO LING

All Papers (Default)

1128/01

1128/02

6091/01

6092/01

Default Accommodation Policy

Applied to all papers unless individually customized in the tabs above.

Apply to All (Reset Overrides)

EXTRA TIME ALLOWANCE

None (0%)

+15%

+25%

+50%

☐ Requires Separate Room

Allocates candidate into an AA-designated quiet venue for written exams (automatically rejoins main cohort for Listening Comprehension).

☐ Preferential / Front-Row Seating

Places student in Row 1 nearest the audio playback source for Listening Comprehension, or near room exit for mobility accommodations.

ACCOMMODATION REMARKS

e.g. Needs large print script, hearing aid, diabetic kit

1 custom paper override(s) active

Cancel

Save Arrangements

Figure: Stage 1 Granular Paper-Level AA Configuration Modal

Step 2: Stage 2 — Examination Timetable & Day Management

2.1 Timetable Ingestion

- Switch to **Stage 2: Timetable**.
- Drag and drop the **official SEAB Timetable PDF** (e.g., 2026_0_Level1_Timetable.pdf) or spreadsheet.
- Plexo automatically parses dates, paper titles, start times, and durations.

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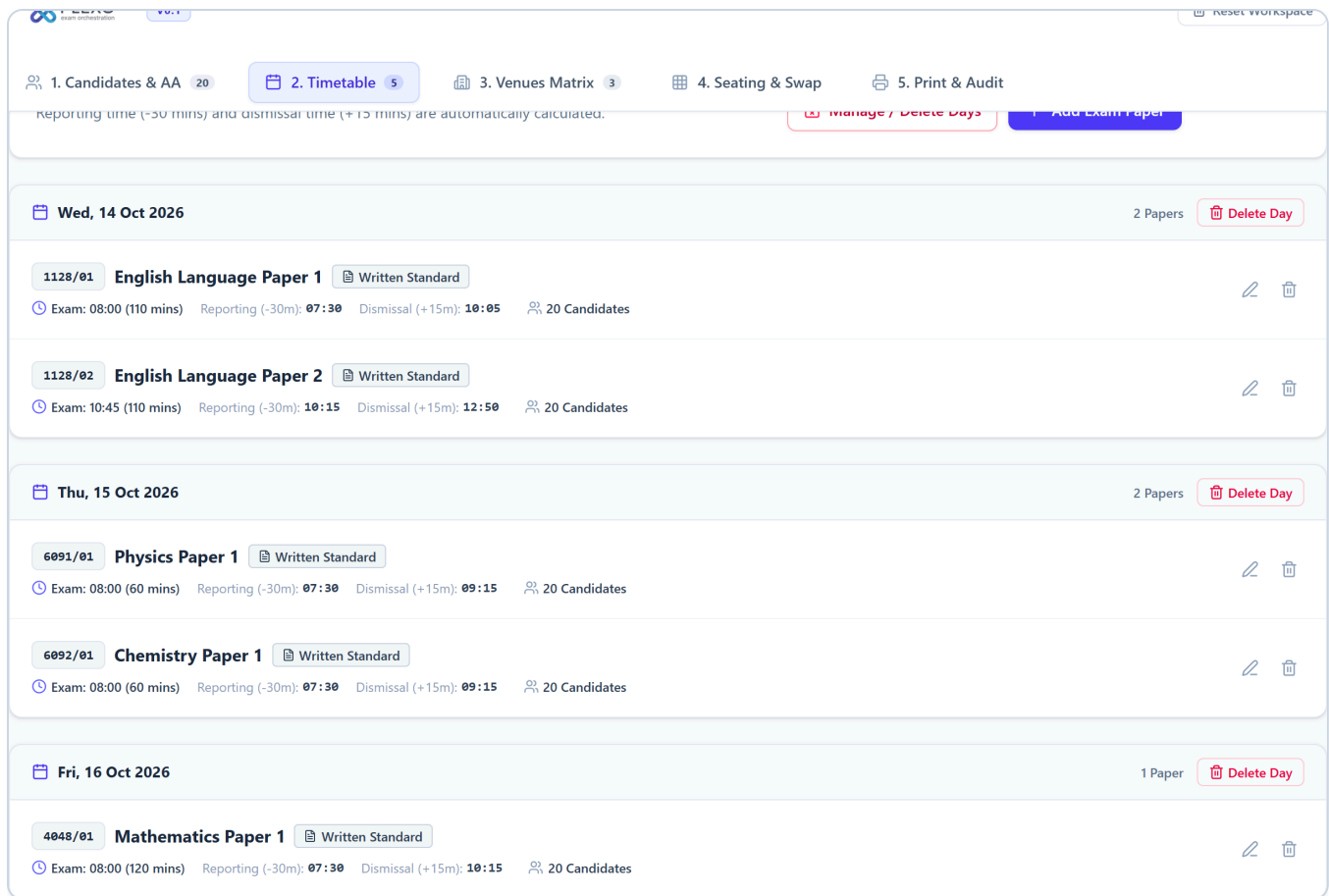


Figure: Stage 2 Examination Timetable Calendar Cards

2.2 Pruning & Day Management

National timetable PDFs often list 200+ subjects nationwide. To keep your schedule clean:

- **Prune 0-Candidate Papers:** Click "**Prune Unenrolled Papers**" to instantly remove papers where your school has zero registered candidates.
- **Delete Specific Days:** Click "**Delete Day**" on any calendar card, or use the "**Manage / Delete Days**" drawer to multi-select and purge irrelevant dates in one click.

Step 3: Stage 3 — Examination Venues Matrix

3.1 Defining School Exam Rooms

- Switch to **Stage 3: Venues Matrix**.
- Review default venues or click "+ **Add Venue**" to add School Halls, Classrooms, Science Labs, or AA Separate Rooms.
- Set rows, columns, PC terminal counts, and audio capability flags.

3.2 Interactive Desk Grid Customization

Every examination room has physical obstacles (pillars, aisles, unusable desks, invigilator pathways).

- Click any desk cell (R1C1 , R1C2 , etc.) to **toggle its active status**.
- Deactivated desks turn grey with a **Ban** icon (Aisle / Pillar) and are skipped by the seating engine.

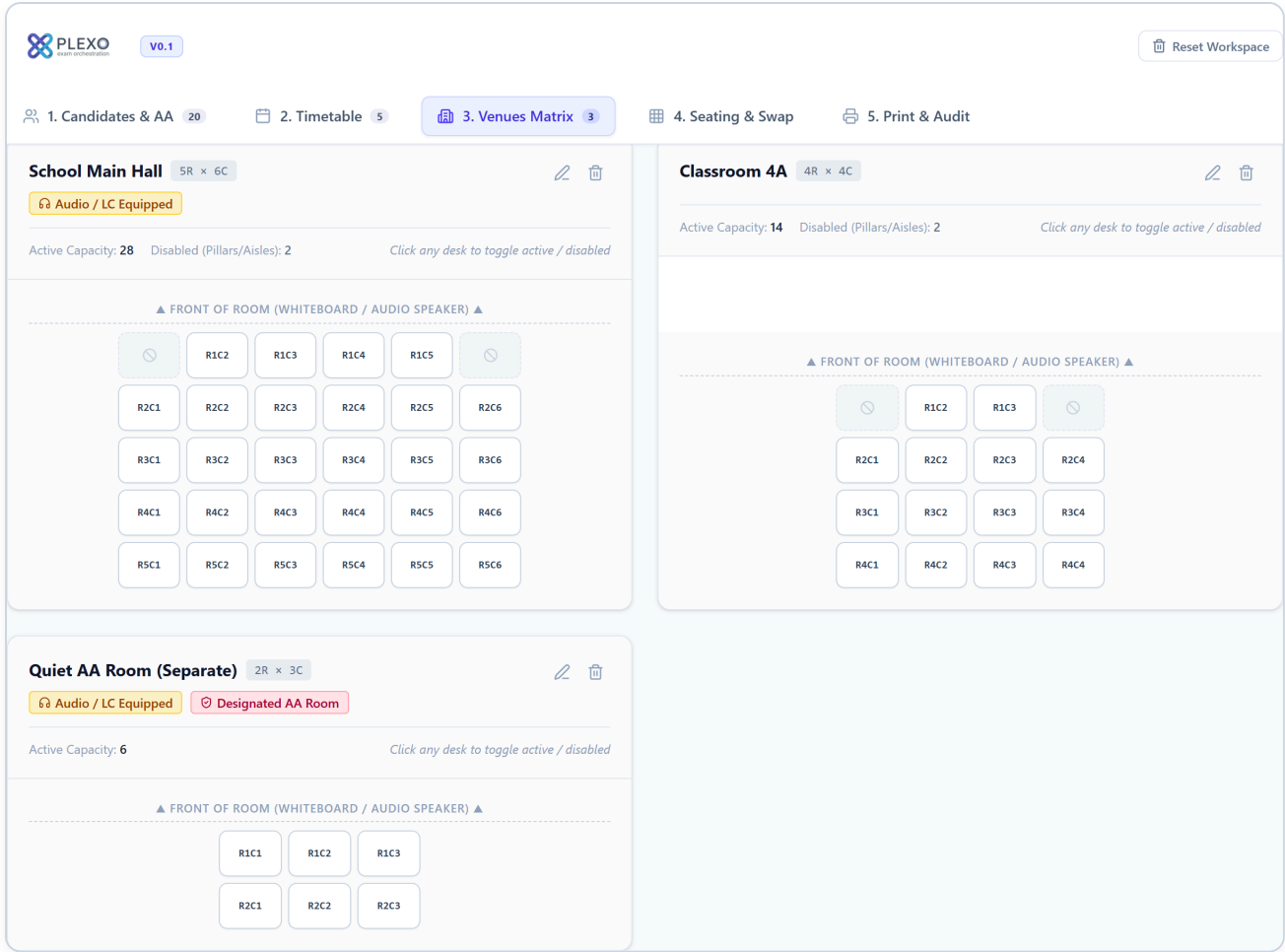


Figure: Stage 3 Examination Venues Matrix & Interactive Grid

Step 4: Stage 4 — Seating Generation & Seat Swapping

4.1 Automated Deterministic Allocation ("Batch Run All")

- Switch to **Stage 4: Seating & Swap**.
- To allocate a single paper, select it from the dropdown and click **"Generate Allocation"**.
- To allocate the entire exam session at once, click **"Batch Run All"**:
 - Runs chronological allocation enforcing **desk continuity** for back-to-back papers.
 - Respects Separate Room segregation and Front-Row Listening Comprehension seating.
 - Displays a cohort summary modal with seated candidate tallies and room notices.

4.2 Interactive 2D Seating Floor Plan

- Visualizes the room layout mirroring the physical hall, with a top banner indicating the **Teacher's Desk / Stage (Front of Room)**.
- Desks display the Candidate Index Number, Name, Desk Label, Shift index, and AA tags.

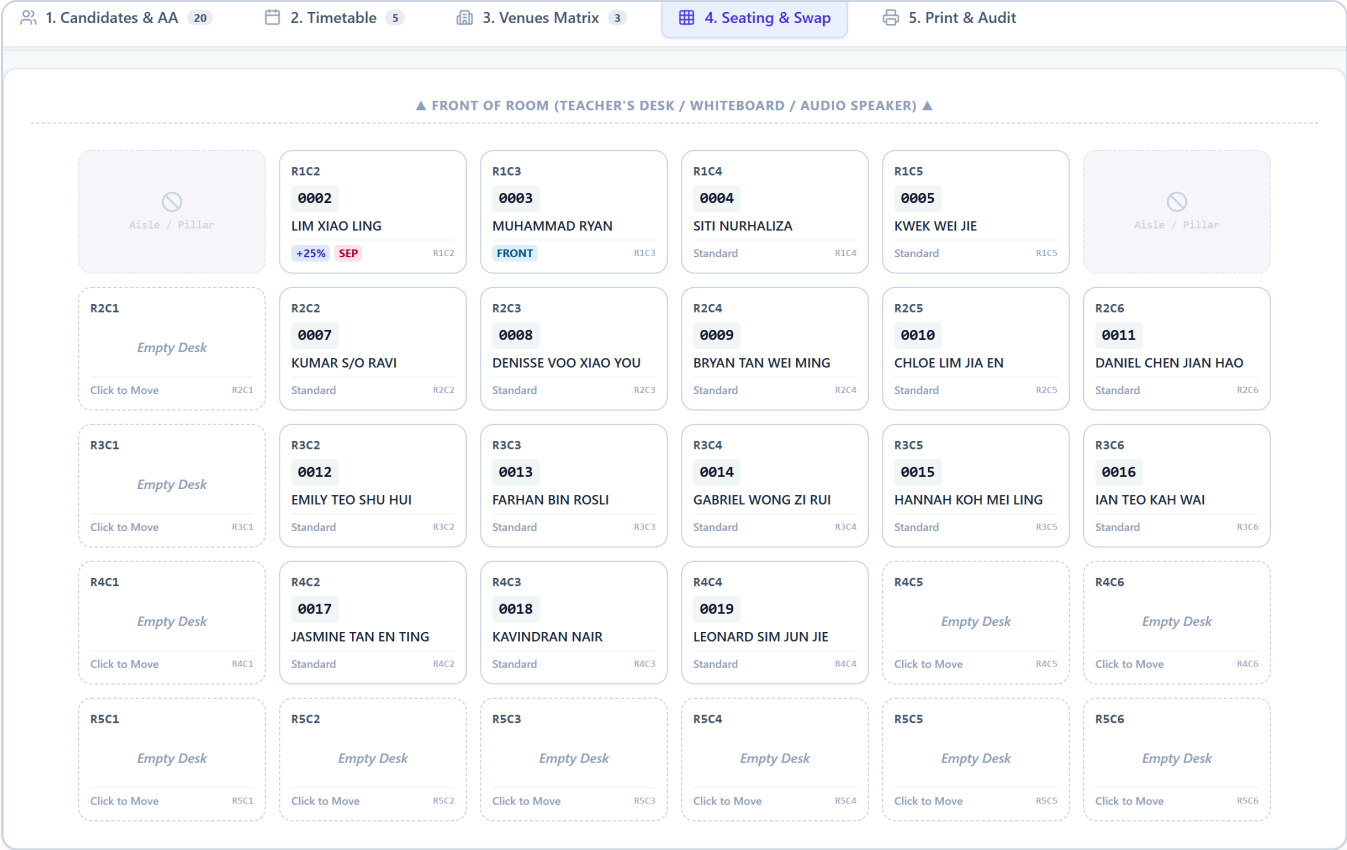


Figure: Stage 4 Interactive 2D Seating Floor Plan Visualizer

4.3 Cross-Room Swapping & Moving to Empty Standby Rooms

- **Swapping Seats Within the Same Room:** Click candidate A, then click candidate B \$to\$ seats swap immediately.
- **Cross-Room Swapping:** Click candidate A in Venue 1 \$to\$ switch room tabs to Venue 2 \$to\$ click candidate B \$to\$ candidates swap across venues with real-time toast confirmation.
- **Moving to an Empty Standby Room:** Click candidate A \$to\$ click "+ Show Empty Rooms" \$to\$ select the empty venue tab \$to\$ click any empty desk to relocate. Alternatively, select the room directly from the "Quick Move: Choose target room..." dropdown.

Reset Workspa

Figure: Stage 4 Cross-Room Seat Relocation Action Toolbar

Step 5: Stage 5 — Print & Audit (SEAB Official Reports)

Switch to **Stage 5: Print & Audit** to access 4 purpose-built audit and operational documents:

5.1 Report 1: Room Use Overview (by Day)

- **Horizontal Axis (Time of Day):** Papers starting at the same time slot (e.g., 08:00) are consolidated into unified time columns with the full time window (e.g., 08:00 – 10:00).
- **Vertical Axis (Venues):** Lists all rooms with room capacities.
- **Matrix Cells:** Displays paper code, paper title, and candidature count in that room for that time slot.

- **Summary Totals:** Shows slot totals and daily room throughput.
- **Excel Export:** Click **"Export Excel"** for an official scheduling matrix spreadsheet.

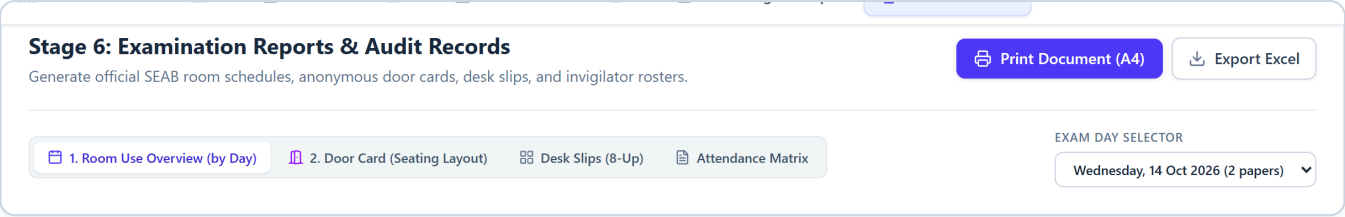


Figure: Stage 5 Report 1: Room Use Overview Schedule

5.2 Report 2: Candidate Door Notice Card (Anonymous Physical Seating)

- Posted on examination room entrance doors for student reference.
- **Strict Privacy Compliance:** Displays **ONLY Candidate 4-Digit Index Numbers** (0001 , 0005 , etc.). Statutory student names are **strictly suppressed**.
- **Header Banner:** Features the prominent **▲ TEACHER'S BENCH (FRONT OF EXAMINATION ROOM) ▲** notice.
- **Multi-Room Page Breaks:** Selecting "All Allocated Venues" automatically formats clean page breaks (A4) between rooms for one-click batch printing.

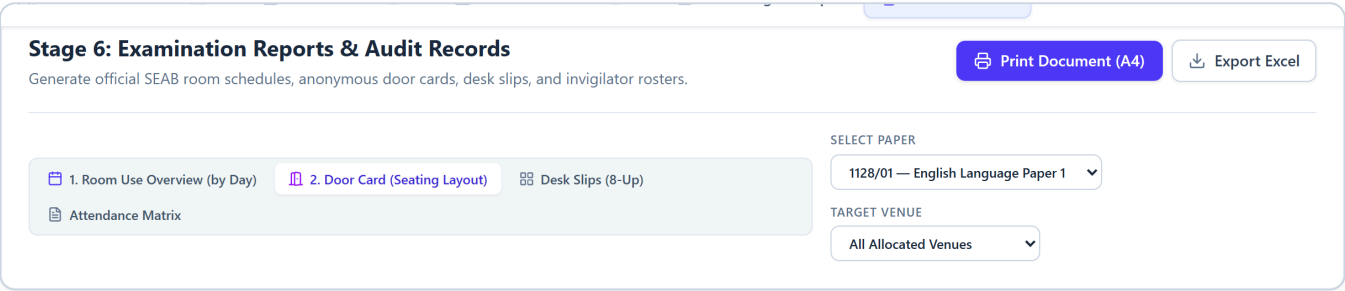


Figure: Stage 5 Report 2: Candidate Door Notice Card

5.3 Report 3: Desk Slips (8-Up Grid on A4)

- Formatted with dashed cut lines for invigilators to place on desks prior to candidate entry.
- Includes Index Number, Candidate Full Name, Desk Label, Paper Code, and Access Arrangement Badges (e.g., +25% EXTRA TIME , SEPARATE ROOM).

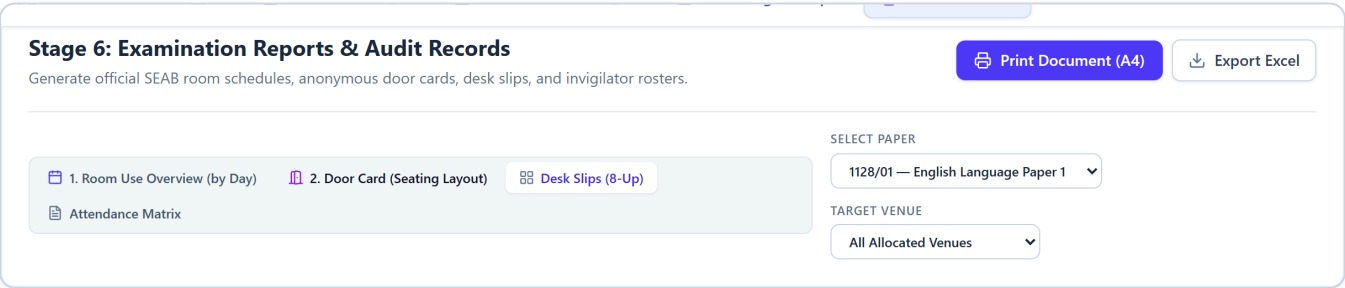


Figure: Stage 5 Report 3: Candidate Desk Slips (8-Up)

5.4 Report 4: Invigilator Attendance & Script Verification Matrix

- Complete candidate roster sorted by desk label.
- Features verification checkboxes for **Absent []**, **Script Collected []**, and candidate signature lines.

- Includes script tally summary footer with Chief Invigilator sign-off.

4. Operational Checklist & Best Practices

1. Before the Exam Season:

- ☐ Double-click `Plexo-v0.1-Standalone.html` to ensure Edge/Chrome launches cleanly.
- ☐ Ingest candidate lists; verify AA counts match internal school Special Educational Needs (SEN) records.
- ☐ Ingest timetable; click "**Prune Unenrolled Papers**" to clean national calendars.
- ☐ Calibrate room seat grids in **Stage 3** to account for pillars, aisles, and AV speaker positions.

2. Generating Seating:

- ☐ Click "**Batch Run All**" in **Stage 4** to execute deterministic allocation with venue continuity.
- ☐ Inspect Separate Rooms to ensure AA students are segregated as required.
- ☐ Use **Cross-Room Swap** or **Quick Move** for any ad-hoc administrative adjustments.

3. Printing & Distribution:

- ☐ Export **Room Use Overview** to Excel for the Operations Head and Facilities team.
- ☐ Print **Door Cards** (A4) to mount outside each exam room.
- ☐ Print and slice **Desk Slips (8-Up)** for table placement.
- ☐ Print **Invigilator Matrices** for chief invigilators and script tallying.

4. Data Purge / Next Exam Cycle:

- ☐ Click "**Reset Workspace**" in the top right header to wipe all data from the browser sandbox when the examination season concludes.